

Per-(perturbation, gene) standard errors of the logFC estimates

How `ComputeSE.py` produces the `se` column fed to `ashr`

1. Setup, per gene g

For a single perturbation p we have two groups of cells in the `AnnData` fed to `ComputeSE.py`:

- **Perturbed cells** (`target_gene == p`): n_p cells with log1p-normalized expression values $x_1^{(p)}, \dots, x_{n_p}^{(p)}$ taken from `adata.X`.
- **Control cells** (`target_gene == non-targeting`): n_c cells with values $x_1^{(c)}, \dots, x_{n_c}^{(c)}$ from the same matrix.

The estimand is the difference of population means on the log1p scale,

$$\beta_{p,g} = \mu_g^{(p)} - \mu_g^{(c)},$$

estimated by the corresponding sample means,

$$\hat{\beta}_{p,g} = \bar{x}^{(p)} - \bar{x}^{(c)}.$$

2. Variance of the estimator

Treating cells as **independent draws** within each group, the sampling variance of each group mean is

$$\text{Var}(\bar{x}^{(p)}) = \frac{\sigma_g^{(p)2}}{n_p}, \quad \text{Var}(\bar{x}^{(c)}) = \frac{\sigma_g^{(c)2}}{n_c}.$$

The two group means are computed from **disjoint sets of cells**, so $\bar{x}^{(p)}$ and $\bar{x}^{(c)}$ are independent. For independent random variables variances add — regardless of whether we are summing or subtracting:

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) \quad \text{when } X \perp\!\!\!\perp Y.$$

There is no minus sign on the right; the sign of the linear combination does not matter once the terms are independent. Applying this with $X = \bar{x}^{(p)}$ and $Y = \bar{x}^{(c)}$ gives the variance of the estimator:

$$\text{Var}(\hat{\beta}_{p,g}) = \text{Var}(\bar{x}^{(p)}) + \text{Var}(\bar{x}^{(c)}) = \frac{\sigma_g^{(p)2}}{n_p} + \frac{\sigma_g^{(c)2}}{n_c}.$$

This is the **Welch / Behrens–Fisher** variance for a two-sample mean difference — it does **not** assume equal variances between groups.

Two sanity-check corollaries:

- *Why the SE shrinks with n .* Each group mean’s variance has n in the denominator. Doubling either group’s cell count roughly halves that group’s contribution to the variance.
- *What can break this formula.* If cells within a group are **not** independent (e.g. they share a library, batch, clone, or cell-cycle state), then $\text{Var}(\bar{x}^{(p)}) > \sigma_g^{(p)2}/n_p$ because the *effective* sample size is smaller than n . The resulting SE is **anti-conservative** — see §5.

3. Estimating σ_g^2

The unknown population variances are replaced by the **unbiased sample variances** (Bessel correction):

$$\hat{\sigma}_g^{(p)2} = \frac{1}{n_p - 1} \sum_{i=1}^{n_p} (x_i^{(p)} - \bar{x}^{(p)})^2,$$

and analogously for the controls. `ComputeSE.py` computes this in a streaming, sparse-aware way without ever materialising the centered matrix. It carries column-wise sums $S_g = \sum_i x_i$ and sums of squares $T_g = \sum_i x_i^2$:

```
s, ss = _sum_and_sumsq(X[group_mask])    # one pass over the sparse matrix
```

A closed-form identity then gives the variance:

$$\hat{\sigma}_g^2 = \frac{n}{n-1} \left(\frac{T_g}{n} - \bar{x}^2 \right) = \frac{n}{n-1} \cdot \underbrace{(\overline{x^2} - \bar{x}^2)}_{\text{population variance}}.$$

In code:

```
ex2      = ss / n
var_pop  = np.maximum(ex2 - mean * mean, 0.0)
var      = var_pop * (n / (n - 1))    # Bessel correction + unbiased
```

The `np.maximum(\ldots, 0.0)` is a numerical guard: with `float32` input and large means, $\overline{x^2} - \bar{x}^2$ can be a tiny negative number because of catastrophic cancellation. Clamp it to zero rather than let `np.sqrt` later return a NaN.

4. Final SE formula

After computing the group means and unbiased variances, the per-(perturbation, gene) standard error is the square root of the Welch variance:

$$\text{SE}(\hat{\beta}_{p,g}) = \sqrt{\frac{\hat{\sigma}_g^{(p)2}}{n_p} + \frac{\hat{\sigma}_g^{(c)2}}{n_c}}.$$

```
mean_diff = mean_p - mean_c
se         = np.sqrt(var_p / n_p + var_c / n_ctrl)
```

This $(\hat{\beta}, \text{SE})$ pair per (pert, gene) is exactly what `ashr::ash(betahat, sebetahat)` consumes downstream to produce the posterior-mean shrunken estimates and `lfsr`.

5. Assumptions and where they fail

Assumption	Holds in our data?
Cells within a group are i.i.d.	Strongly violated. Cells share library / batch / cell-cycle / clonal structure, so the effective sample size is $< n$. The SE is therefore anti-conservative — true uncertainty is larger than computed.
Sample means are approximately Gaussian (CLT)	Roughly OK for moderately-expressed genes when n is in the hundreds. Breaks for very small n or very sparse genes.
$\hat{\sigma}_g^2$ is a reliable estimate of σ_g^2	OK at large n . Breaks at small n: the sample variance can be exactly 0 just because all cells in a small group happen to have the same value at a sparse gene — the SE then collapses to ≈ 0 and ashr treats the estimate as perfectly precise. This is the failure mode behind the small- n posterior-mean inflation observed in the DMSO replicate diagnostic.
Mean difference on the log1p scale $\approx$ log fold-change	Approximate. For highly-expressed genes $\log1p(x) \approx \log(x)$, so the difference is in natural-log units . For low-count genes log1p departs from log appreciably.
Perturbed and control cells are independent	True — no cell is in both groups.

6. In one sentence

We compute, gene by gene, $\hat{\beta} = \bar{x}_{\text{pert}} - \bar{x}_{\text{ctrl}}$ and $\text{SE} = \sqrt{\hat{\sigma}_{\text{pert}}^2/n_{\text{pert}} + \hat{\sigma}_{\text{ctrl}}^2/n_{\text{ctrl}}}$ — the textbook Welch SE for a two-sample mean difference — on log1p-normalized expression, treating each cell as one independent observation.